

could also be a range of applications in battlefield weapons systems.

Contactless battery charging

The world's first true single-chip contactless power transmitter accompanied by the industry's highest-output-power single-chip receiver solution has been developed. The technology, developed by a California-based analogue and digital company, IDT, may feature in future smartphones.

The device has to be placed on a specific spot, where it charges via inductive coupling. A current passes through the electric coil inbuilt in the transmitter, which creates an electromagnetic field. A similar coil on the back of the phone senses the electromagnetic field and turns it into electricity whenever the phone is in the range. The chip can sense different phone models. While wireless charging needs a charger to be attached to the sockets via a wire, contactless charging needs no wires. The solution consolidates the function of nine ICs into one.

Modified bacteria generate electricity from sewage

A fuel cell powered by naturally occurring bacteria has successfully converted 13 per cent of the energy in sewage into electricity, and cleaned the waste water at the same time. It uses naturally occurring microbes to break down organic waste and produce electrons and protons. The electrons are collected by an anode, while the protons pass through a permeable membrane to a cathode. The resulting voltage between the two electrodes enables the fuel cell to produce an electric current.

Says researcher Orianna Bretschger of the J. Craig Venter Institute in San Diego, California, "If we could recover the energy, we could do waste water treatment for free."

By genetically modifying microbes to enhance their ability to consume organic waste and better shuttling of electrons to an electrode, Bretschger hopes to harvest 30 to 40 per cent of the available energy.



IDT's wireless recharging chips (on right) versus a rival product

Ultra-thin solar cells developed

Researchers from Johannes Kepler University of Austria and University of Tokyo have developed solar cells that are thinner than a thread of spider silk. These cells are said to be flexible enough to wrap around a single human hair.

Being extremely thin, light and flexible, they can be used in portable electrical charging devices or wearable electronics clothing, the journal *Nature Communications* reports. The thin-film device is about 1.9 micro-metres thick. It comprises electrodes on a plastic foil.

One of the researchers, Tsuyoshi Sekitani from the University of Tokyo, explains, "Being ultra-thin means you don't feel its weight and it is elastic. You could attach the device to your clothes like a badge to collect electricity (from the sun). Elderly people who might want to wear sensors to monitor their health would not need to carry around batteries."

The researchers expect to put the technology to practical use in the next five years.

100T magnetic field created

A team of engineers has created a 100-Tesla magnetic field, which is about two million times powerful than the earth's magnetic field. Engineers from Los Alamos National Laboratory

have developed this magnanimous magnetic field in a period of 15 years.

Chuck Mielke, director, Pulsed Field Facility at Los Alamos, said, "There are higher magnetic fields that have been produced but the magnets used in them destruct themselves. The magnets used by our team are first-of-their-kind in the world that work non-destructively at 100T intensity on a regular basis."

The team employed a 100T pulsed multishot magnet, which is a combination of seven coil sets. These coils are powered by a mega 1200MJ generator. The engineers are now working to use these magnetic fields to study upper critical fields of superconductors, electrical resistivity, quantum magnetic transitions, fibre-optic dilatometry and optical spectroscopy.

These magnets will be used to understand the behavioural patterns of various materials when subjected to intense magnetic fields. This study of behaviour of various materials at very low temperatures and very high magnetic fields can be applied to enhance the knowledge of superconductivity.

World's first quantum network

Researchers at Max Planck Institute of Quantum Physics (MPQ) have successfully developed a quantum network